

vitals_by_patient

partition key: *patient_id*
clustering: *measured_at DESC*

Q1: a patient's recent readings

ward_risk_snapshot

partition key: *ward_id, scorer_version*
clustering: *risk_score DESC, patient_id*
TTL 120 s

Q2: the ward list, worst first

risk_scores_by_patient

partition key: *patient_id, scorer_version*
clustering: *scored_at DESC*

Q3: a patient's score history,
per rule version

alerts_by_ward

partition key: *ward_id, sim_date, scorer_version*
clustering: *alert_time DESC, alert_id*
TTL 3 h

Q4: today's alerts on the ward

labs_by_patient

partition key: *patient_id*
clustering: *test_type, collected_at DESC*

Q5: a patient's lab results

daily_patient_summary

partition key: *ward_id, sim_date*
clustering: *patient_id, scorer_version*

Q6: the daily PDF report

Each table is shaped for exactly one question, so every read touches one partition and none needs ALLOW FILTERING. Rows that go stale expire by TTL instead of DELETE, which would leave tombstones behind.